Data Communication Kya Hai?

- Data Communication ka asan matlab hai kisi bhi transmission medium ke zariye do devices ke darmiyan data ka tabadla karna.

3. Data Communication Ke Components

Data bhejte waqt paanch (5) cheezein zaroori hoti hain:

- **Message:** Yeh wo asal data ya information hai jo bheji ja rahi hai (jaise text, audio, ya video).
- **Sender:** Wo device jo data bhejti hai (jaise computer ya mobile phone).
- **Receiver:** Wo device jo message ko wasool karti hai (jaise TV ya doosra phone).
- **Transmission Medium:** Yeh wo rasta hai jis ke zariye data safar karta hai, jaise cables ya radio waves.
- **Protocol:** Yeh communication ko control karne ke liye banaye gaye rules hain. Agar yeh rules na hon

toh accidents hote hain, bilkul traffic signals ya WhatsApp ki chatting ke rules ki tarah.

4. Ek Achi Communication Ki Khasoosiyat (Characteristics)

Data communication ki karkardagi (effectiveness) in 4 cheezon par depend karti hai:

- **Delivery:** Data bilkul sahi destination (manzil) par pohanchna chahiye. Jaise agar email Ali ko bheji aur Ahmed ko mil jaye toh delivery fail ho jati hai.
- **Accuracy:** Data bilkul waisa hi receive hona chahiye jaisa bheja gaya tha. Agar aap ne $500 bhejein aur receive karne wale ko sirf $50 milein, toh accuracy kharab hai.
- **Timeliness:** Data waqt par deliver hona zaroori hai. Jaise live class mein agar video dair se aaye toh lecture samajhna mushkil ho jata hai.
- **Jitter:** Iska matlab hai data ke aane ki speed ka uneven hona (kabhi fast, kabhi slow). Voice ya video call ke dauran awaz ka katna ya video rukhna Jitter ki wajah se hota hai.

5. Data Representation (Data ko computer mein dikhana)

Computer har qism ke data ko apne samajhne ke laiq (0s aur 1s / binary) mein convert karta hai. Data mukhtalif types ka hota hai:

- **Text:** Ise bits (0s ya 1s) ke pattern mein ASCII ya Unicode ke zariye likha jata hai.
- **Numbers:** Yeh binary, decimal, ya floating-point format mein represent kiye jate hain.
- **Images:** Tasweerein chhote chhote dots (pixels) ko mila kar banti hain. Har pixel ki apni color value (RGB) hoti hai.
- **Audio/Video:** Audio lagataar (continuous) awaz hoti hai. Jabke video asal mein bohat sari tasweeron (frames) aur awaz ka ek sequence hoti hai jisse motion (harkat) ka ehsas hota hai.

6. Data Flow

Data flow ka matlab hai ke data kis direction (samt) mein safar kar raha hai. Iski 3 types hoti hain:

- **Simplex:** Data sirf aur sirf ek direction mein jata hai. Bhejne wala sirf bhejta hai aur wasool karne wala wapas kuch bhej nahi sakta. Misal ke tor par, Keyboard se Computer mein data jana.

- **Half-Duplex:** Is mein data dono directions mein ja sakta hai, lekin ek waqt mein sirf ek device data bhej sakti hai. Jaise Walkie-Talkie par jab ek insan bolta hai toh doosra sirf sunta hai.

- **Full-Duplex:** Dono devices ek hi waqt mein aik sath data bhej aur wasool kar sakti hain. Iski sab se achi misal Telephone Call hai, jahan dono log ek sath bol aur sun sakte hain.

---

1. Transmission Media Kya Hai?

- Transmission medium ek physical rasta ya connection (bridge) hota hai jis ke zariye data ek device se doosri device tak safar karta hai.

- **Transmitter:** Wo device jo data bhejti hai (jaise message bhejta hua mobile).

- **Receiver:** Wo device jo data wasool karti hai.

Transmission media ki do (2) bari iqsaam (categories) hain: **Guided Media** (Taar wali) aur **Unguided Media** (Baghair taar wali / Wireless).

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2. Guided Media (Wired / Taar Wali Communication)

- Ise Bounded transmission media bhi kehte hain kyunke isme data taaron (physical links) ke andar qaid ho kar jata hai.

- Yeh fast, secure hoti hai aur amuman chhote faslon (shorter distances) ke liye istemal hoti hai.

- Iski teen (3) aham iqsaam hain:

A. Twisted Pair Cable

- Isme do insulated taaron ko aapas mein bal (twist) diya jata hai aur yeh sab se zyada istemal hone wali media hai.

- Yeh LANs (Local Area Networks), landline telephones aur ethernet mein aam istemal hoti hai.

- Iski do (2) types hain:
  - **Unshielded Twisted Pair (UTP):** Isme taaron ke upar koi extra metallic cover nahi hota, bas twist karne se interference kam ki jati hai. Yeh sasti aur lagane mein asan hoti hai, lekin iski capacity STP se kam hoti hai aur yeh short distance ke liye hai.
  - **Shielded Twisted Pair (STP):** Isme taaron ke upar ek metallic shield hoti hai jo shor (noise) se bachati hai. Yeh UTP se zyada fast aur behtar performance deti hai, lekin lagane mein mushkil aur mehngi hoti hai. Yeh banks aur industrial areas mein istemal hoti hai.

B. Coaxial Cable

- Isme ek markazi (central) copper wire hoti hai, jiske upar insulation aur ek metal shield hoti hai jo shor (noise) se bachati hai.
- Yeh twisted pair cable se behtar noise protection deti hai.

- **Fayde:** Yeh sasti hoti hai, lagane mein asan hai, aur iski bandwidth zyada hoti hai.
- **Nuqsan:** Agar ek cable kharab ho jaye toh poora network ruk sakta hai.
- **Misaal:** Cable TV aur CCTV cameras ki wiring mein yahi istemal hoti hai.

C. Optical Fibers

- Yeh shishay (glass) ya plastic se banti hai aur data ko light (roshni) ke reflection ke zariye bhejti hai.
- Iske andar ek "Core" hota hai jis mein light safar karti hai, aur uske bahar "Cladding" hoti hai jo light ko bahar nikalne se rokti hai.
- Yeh lamba fasla tay karne aur bohat zyada data, tezi se bhejne ke liye behtareen hai.
- **Misaal:** PTCL ka fiber internet aur hospital/university ke high-speed networks.

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3. Unguided Media (Wireless / Baghair Taar Wali)

- Ise Unbounded media bhi kehte hain kyunke isme kisi physical taar ki zaroorat nahi hoti; signal seedha hawa (air) ke zariye broadcast hota hai.
- Yeh lambay faslon (larger distances) ke liye istemal hoti hai.
- Iski teen (3) major types hain:

A. Radio waves

- Yeh hawa ke zariye data bhejti hain aur inhein taaron ki zaroorat nahi hoti.
- Ek transmitter radio waves bhejta hai aur receiver unhein hawa se catch karta hai.
- **Misaal:** FM radio, Wi-Fi, aur Bluetooth.

B. Microwaves

- Isme signals seedhi line (line of sight) mein safar karte hain, is liye bhejne aur wasool karne walay antennas ka aamne-samne hona zaroori hai.

- Signal kitni door jaye ga, iska inhisar antenna ki height par hota hai.
- **Misaal:** Mobile phone ke towers aur satellite communication.

C. Infrared

- Yeh bohat hi chhote faslay (very short distance) par data bhejne ke liye istemal hoti hai.
- Isme bhi signal seedhi line mein jata hai aur darmiyan mein koi rukawat (jaise deewar) nahi honi chahiye (line of sight).
- **Misaal:** TV ka remote control, wireless headphones, aur phone ke darmiyan file transfer.

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1. Computer Network Kya Hai?

- Computer network mukhtalif devices (jaise computers, servers, smartphones, aur printers) ka ek aisa system hai jo aapas mein connect ho kar data exchange karte hain.
- Yeh wired (taaron wale) ya wireless ho sakte hain.

- **Misaal:** School ki lab mein sab computers ka ek hi printer se connect hona, ya ghar mein sab mobiles ka ek Wi-Fi se connect hona.

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2. Network Criteria (Network ko judge karne ke mayaar)

Ek network kitna acha kaam kar raha hai, isko 3 cheezon se parkha jata hai:

- **Performance:** Data kitni tezi se bheja aur wasool kiya ja raha hai. Is mein speed aur delay waghera shamil hain. (Misaal: Agar aap ki online class ki video bina ruke chal rahi hai toh performance achi hai) .

- **Reliability:** Network kitna qabil-e-aitmaad hai aur yeh kitni dafa kharab hota hai. (Misaal: Agar university ka Wi-Fi poora din bina disconnect hue chale, toh wo reliable hai) .

- **Security:** Data ko un logon se bachana jin ke paas permission nahi hai. Is mein passwords, firewalls aur encryption shamil hain.

3. Connections Ki Iqsaam (Types of Connection)

- **Point-to-Point:** Sirf do (2) devices ke darmiyan direct connection. Jaise ek computer ka direct cable ke zariye printer se connect hona.
- **Multipoint:** Jab do se zyada devices ek hi communication link share karti hain. Jaise bohat se students ka ek hi Wi-Fi router istemal karna ya lab mein sab PCs ka ek switch se connect hona.

4. Network Topology (Network ka Design/Layout)

Network mein computers aur cables ko jis design mein lagaya jata hai, use topology kehte hain. Iski aham iqsaam yeh hain:

- **Star Topology:** Sab devices ek central "Hub" ya switch se connect hoti hain. Ise lagana asan hai aur ek taar kharab hone se baqi network par asar nahi parta. Lekin agar main Hub kharab ho jaye toh poora network band ho jata hai.

- **Ring Topology:** Devices ek circle (golai) ki shakal mein connect hoti hain aur data ek hi direction mein chalta hai. Iska nuqsan yeh hai ke kisi ek device ke kharab hone se poora network ruk jata hai.

- **Bus Topology:** Sab devices ek hi main lambi cable se connect hoti hain. Yeh kafi sasta aur asan hai, lekin agar main taar toot jaye toh poora network fail ho jata hai.

- **Mesh Topology:** Har device network mein majood bohat si doosri devices se direct connect hoti hai. Yeh bohat reliable hai (agar ek rasta kharab ho jaye toh data doosre raste se chala jata hai), lekin iski setting complex aur mehngi hoti hai.

- **Hybrid Topology:** Yeh do ya do se zyada topologies (jaise Star aur Bus) ko mila kar banai jati hai. Yeh badi jaghon ke liye achi hai lekin iska setup mehnga aur mushkil hota hai.

- **P2P (Peer-to-Peer):** Is mein sab computers barabar hote hain aur kisi central server ke baghair direct aapas mein data share karte hain. Jaise 2 laptops ko LAN cable se connect kar ke file share karna. Yeh sasta hai lekin is mein security kam hoti hai.

- **Daisy Chain Topology:** Is mein devices ek line mein, ek ke baad ek (series mein) connect hoti hain. Yeh design simple hai lekin agar ek device fail ho toh aage wala connection bhi ruk jata hai.

Computer Network Kya Hai?

- Ek computer network aesa system hai jahan mukhtalif devices (jaise computers, servers, smartphones, aur printers) ek doosre se data exchange karne ke liye connect hote hain.

- Yeh connections taaron ke zariye (wired) ya baghair taar ke (wireless) ho sakte hain.

- Yeh chote home networks se le kar poori duniya mein phaile Internet tak ho sakte hain.

- **Aam Misalein:**
    - School ki lab mein sab computers ka ek hi printer se connect hona.

    - Ghar ke sab mobile phones ka ek hi Wi-Fi se connect hona.

    - Bank ke ATMs ka main server se connect hona.
    Yeh sab computer networks ki misalein hain.

Network Ki Iqsaam (Types of Networks)

1. Personal Area Network (PAN)

- Yeh ek bohat chhota network hota hai jo sirf ek shakhs apni personal devices ko connect karne ke liye use karta hai.

- Iski range sirf kuch meters tak (jaise ek kamre ke andar) hoti hai.

- Isme aam taur par ek device baaki devices ko control karti hai aur zyada tar Bluetooth ka istemal hota hai.

- **PAN ki Do Iqsaam (Types):**

    - **Wired PAN:** Jab devices data cables (taaron) ke zariye connect hon. Data taaron ke zariye transfer hota hai aur connection kafi stable hota hai. Misal: Phone ko data cable se laptop se connect karna.

    - **Wireless PAN (WPAN):** Jab devices baghair kisi cable ke radio waves (jaise Bluetooth) ke zariye connect hon. Misal: Phone ke sath Bluetooth earbuds ya smartwatch connect karna.

2. Local Area Network (LAN)

- LAN ek chote area mein computers ko aapas mein jorta hai, jaise apka ghar, office, school, ya lab.

- Iski range ek building ya campus tak mehdood hoti hai.

- Isme devices Ethernet cables ya Wi-Fi ke zariye connect hoti hain, aur data ko manage karne ke liye switch ya router ka istemal hota hai.

- Misal: Ek department ki lab mein sab computers ka aapas mein connect hona.

3. Metropolitan Area Network (MAN)

- MAN mukhtalif LANs ko mila kar ek poore shehar (city) ya baday campus ka network banata hai.

- Isme fiber optic cables ka istemal hota hai aur isay badi organizations manage karti hain.

- Misal: Ek shehar mein university ke mukhtalif campuses ko aapas mein connect karna ya city-wide Wi-Fi.

4. Wide Area Network (WAN)

- WAN sab se bada network hai jo poore mulk ya poori duniya ke networks ko aapas mein connect karta hai.

- Yeh data bhejne ke liye satellites, samundar ke andar bichi cables, fiber optics, aur routers ka istemal karta hai.

- Misal: Duniya bhar mein WhatsApp message bhejna ya Pakistan se Google ke servers ko access karna.

Piconet Kya Hai?

- Piconet ek bohat chhota wireless network hai jo Bluetooth ki madad se banta hai.

- Isme ek main device hoti hai jise **Master** kehte hain, aur baaki devices ko **Slave** kehte hain.

- **Master Device:** Yeh baat-cheet (communication) ko control karti hai aur faisla karti hai ke kon kab data bhejega.
- **Slave Devices:** Yeh Master ki di gayi instructions ko follow karti hain.
- **Rule:** Ek Piconet mein 1 Master aur zyada se zyada 7 Slave devices ho sakti hain.
- Misal: Jab aap apne phone (Master) ke sath earbuds, smartwatch, aur car ka Bluetooth (Slaves) connect karte hain, toh yeh ek Piconet ban jata hai.

Ethernet Aur Iski Iqsaam

- **Ethernet:** Yeh wired LANs (taar wale local networks) mein sab se zyada use hone wali technology hai. Yeh data ko 'frames' ki shakal mein bhejti hai aur devices ki pehchan ke liye MAC addresses use karti hai. Iski basic speed 10 Mbps hoti hai.

- **Fast Ethernet:** Yeh aam Ethernet ka teez (fast) version hai jo chote office networks mein use hota hai. Iska kaam karne ka tareeqa same hai magar speed aam Ethernet se 100 guna behtar hoti hai.

- **Metro Ethernet:** Yeh Ethernet technology ko shehar (city) ke level tak barha deta hai. Yeh fiber optic cables use karta hai jis se speed bohat taiz milti hai. Yeh badi companies apne shehar ke mukhtalif branches ko connect karne ke liye use karti hain.

Muazna (Comparison Table):

Feature	**Ethernet**	**Fast Ethernet**	**Metro Ethernet**
Speed	10 Mbps	100 Mbps	1–10 Gbps
Area	LAN (chhota)	LAN (chhota/darmiyana)	City / MAN
Cable	Copper (taanba)	Copper (taanba)	Fiber optic

Cost (Kharcha)	Low (Kam)	Medium (Darmiyana)	High (Zyada)
Usage (Istemal)	Puranay LANs mein	Naye/Modern LANs mein	Poore shehar ke networks mein

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Internetwork

- Jab do ya do se zyada networks ko aapas mein jor diya jaye to usay "Internetwork" kehte hain.
- **Internet** duniya ka sab se bada internetwork hai kyunke yeh duniya bhar ke dheron LANs, MANs, aur WANs ko aapas mein milata hai.
- **Misal:** Jab aapka apna ghar ka Wi-Fi network kisi aur network (jaise Google ke servers) se connect hota hai, toh is connection ko internetwork kehte hain.

Protocols aur Standards

- **Protocol Kya Hai?**

    - Protocol un rules (qawaneen) ka ek set hai jo computers ek dusre se baat cheet karne ke liye follow karte hain.

    - Isey aap computers ki "language" (zaban) samajh sakte hain. Jaise insaan English ya Urdu mein baat karte hain, wese hi computers HTTP, TCP, aur IP jaise protocols use karte hain.

    - Bina protocols ke, computers ek dusre ki baat nahi samajh sakte.

- **Standards Kya Hote Hain?**
    - Standards official rules hote hain jo is baat ki tasdeeq karte hain ke mukhtalif companies ki devices aapas mein mil kar kaam kar sakein.

    - **Misal:** Ek HP ka laptop Dell ke server se baat kar sake, ya apka mobile kisi bhi Wi-Fi router se connect ho jaye. Yeh is liye mumkin hai kyunke sab ek hi standard follow karte hain.

    - **Asaan Misal:** Jaise traffic rules mein Red ka matlab rukna aur Green ka matlab chalna hai, sab inhein follow karte hain toh accidents nahi hote.

- **Internet Standards:**
  - Yeh woh rules hain jo khaas taur par Internet ke liye banaye gaye hain.
  - Yeh tay karte hain ke data kaise bheja jaye, websites kaise kaam karein, aur emails kaise deliver hon.
  - **Misalein:** HTTP (web browsing ke liye), SMTP (email bhejne ke liye), aur IP (addressing ke liye). Jab aap google.com kholte hain, toh aapka browser aur server dono internet standards follow kar rahe hote hain.

Layered Tasks (Sender, Carrier, aur Receiver)

Communication ko mukhtalif hisson (layers) mein taqseem kiya jata hai aur har hisse ka apna ek kaam hota hai.

- **Sender (Bhejne Wala):** Yeh data tayar karta hai, usay chote packets mein torta hai, aur address lagata hai. Misal: Jab aap WhatsApp par message bhejte hain.

- **Carrier (Le Kar Jane Wala):** Yeh data ko network ke zariye le kar jata hai, jaise cables, Wi-Fi, ya fiber optics ke zariye. Misal: Internet ki taarein ya Wi-Fi signals.

- **Receiver (Wusool Karne Wala):** Yeh data receive karta hai, packets ko wapas jorta hai, aur user ko message dikhata hai. Misal: Jab aapke dost ko message milta hai aur woh parhta hai.

OSI Model aur Layered Architecture

- **Layered Architecture:** Iska matlab hai ke har layer ek makhsoos kaam karti hai aur doosri layers se azaad (independently) kaam karti hai.

- **OSI Model:** Yeh ek 7-layer ka reference model hai jo sikhata hai ke data sender se receiver tak kaise safar karta hai. Yeh networking ko samajhna asaan banata hai.

  - 7. Application Layer: User interaction, jaise Browser ya Email.

  - 6. Presentation Layer: Data ka format aur uski encryption karna.

  - 5. Session Layer: Session ko manage karna.
  - 4. Transport Layer: Data ka hifazat se pohanchna (Reliable delivery).
  - 3. Network Layer: Rasta tay karna (Routing) aur IP address lagana.
  - 2. Data Link Layer: MAC address lagana aur framing karna.
  - 1. Physical Layer: Cables aur signals ka kaam.

Peer-to-Peer (P2P) Process

- Is network mein dono devices sender aur receiver dono ka kaam kar sakti hain.
- Isme kisi main "boss" ya central server ki zaroorat nahi hoti.
- **Misal:** Bluetooth ke zariye file bhejna ya Torrent se downloading karna. Dono dost aapas mein baat kar sakte hain aur sun bhi sakte hain.

Encapsulation (Data ko Pack Karna)

- **Encapsulation Kya Hai?** Jab data sender se receiver ki taraf jata hai, toh raste mein har layer us data ke upar apna ek cover ya extra information lapet deti (wrap kar deti) hai. Isay layer by layer pack karna kehte hain.

- **Encapsulation ka Safar (Flow):**
  1. **Application Layer (Message):** Asal message banati hai, jaise "Hello".

  2. **Transport Layer (Segment):** Isme port number shamil hota hai aur data ko chote hisson (segments) mein tora jata hai taake aram se bheja ja sake.

  3. **Network Layer (Packet):** Segment ab Packet ban jata hai. Isme sender aur receiver ka IP address lagaya jata hai taake data apna sahi rasta dhoondh sake.

  4. **Data Link Layer (Frame):** Packet ab Frame ban jata hai. Isme devices ka MAC address lagta hai jo local network mein delivery ke kaam aata hai.

5. **Physical Layer (Bits):** Aakhir mein data (Frame) 0s aur 1s yani signals mein badal jata hai aur taaron (cables) ya Wi-Fi ke zariye safar karta hai.

OSI Model Kya Hai?

- OSI (Open Systems Interconnection) Model ek concept ya reference model hai jo network communication ko 7 asaan hisson (layers) mein taqseem karta hai.
- Yeh hardware ya software nahi hai, balke networking ko samajhne, design karne, aur uske masail (problems) ko theek karne ka ek tareeqa hai.
- Isey tarteeb se yaad rakhne ki ek asaan trick yeh hai: **A**ll **P**eople **S**ee **T**o **N**eed **D**ata **P**rocessing.

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7 Layers of OSI Model

7. Application Layer (User Level)

- Yeh layer direct user ke sath interact karti hai aur applications ko network ki sahoolat (services) faraham karti hai.
- **Kaam:** Yeh aapka message leti hai aur usay bhejne ke liye tayar karti hai.

- **Misalein:** Web browser (Chrome), Email (Gmail), WhatsApp.
- **Protocols:** Isme HTTP (websites kholne ke liye), HTTPS (secure websites ke liye), FTP (files download/upload karne ke liye), aur SMTP (email bhejne ke liye) istemal hote hain.

6. Presentation Layer (Translator)

- Yeh layer data ko translate, encrypt aur compress karti hai.
- **Translation (Format Conversion):** Computer insaan ki zaban nahi samajhta, is liye yeh layer text, image aur video ko machine ki zaban (jaise ASCII, JPEG, MP4) mein translate karti hai.
- **Encryption (Security):** Yeh data ko chupa deti hai (jaise "Hello" ko "A7#K9@P$%" mein badal dena) taake hackers usay na parh sakein.
- **Compression (Size Kam Karna):** Yeh data ka size chota karti hai (jaise 5MB ki picture ko 1MB kar dena) taake internet par taizi se bheja ja sake.

5. Session Layer (Session Management)

- Yeh layer do devices ke darmiyan connection (baat-cheet) ko start karti hai, qaim (maintain) rakhti hai, aur khatam (end) karti hai.
- **Misal:** Jaise phone call milana (Start), baat karna (Maintain), aur call kaat dena (End). Agar baat ke

dauran internet chala jaye toh yeh session ko pause karti hai aur wapas aane par khud connect kar leti hai.

4. Transport Layer (Delivery Manager)

- Yeh layer is baat ki tasdeeq karti hai ke data theek aur mukammal taur par pohnch jaye.
- **Kaam:** Yeh baday data ko chote tukron mein torti hai jisey "Segments" ya "Packets" kehte hain. Yeh flow control karti hai aur agar koi packet raste mein ghum ho jaye toh usay wapas mangwati hai (Error Checking).
- **TCP aur UDP: * TCP:** Data bohat ehtiyat se bhejta hai aur ghum hone par wapas bhejta hai (Misal: Email, Bank transfer, File download).
    - **UDP:** Data bohat taizi se bhejta hai magar speed ke liye accuracy qurban kar deta hai, jo data ghum ho jaye usay ignore karta hai (Misal: Live video streaming, Online games).

3. Network Layer (Internet ka GPS / Route Finder)

- Yeh layer tay karti hai ke data ko kis raste se jana chahiye aur sab se chota/fast rasta (route) konsa hai.
- Jis tarah Google Maps rasta dhoondhta hai, bilkul wese hi Network layer "Routers" ka istemal kar ke

best path dhoondhti hai.
- Yeh devices ko pehchanne ke liye **IP Address** (Logical address) lagati hai.

2. Data Link Layer (Local Delivery)

- Yeh layer data ko aapke local network (jaise apke ghar ka WiFi ya office ka LAN) ke andar deliver karti hai.
- Internet (Network Layer) par IP address use hota hai, jabke local network mein yeh **MAC Address** (device ka permanent physical address) use karti hai.
- **Framing:** Yeh data packets ko mazeed chote hisson mein taqseem karti hai jinhein "Frames" kehte hain.
- Iski do (2) sublayers hoti hain: **MAC** (jo decide karti hai kon data bhejega) aur **LLC** (jo is baat ko manage karti hai).

1. Physical Layer (Signals & Hardware)

- Yeh layer data ko physically taaron (cables) ya wireless signals ke zariye bhejti hai.
- Isme koi logic ya address nahi hota, sirf bijli (electrical), roshni (light), ya radio signals (WiFi/Bluetooth) hote hain.
- Yeh frames ko **Bits (0 aur 1)** mein tabdeel kar ke bhejti hai.

- Hub, Repeater, Cables, aur Antenna is layer par kaam karte hain.

Collision Kya Hai? (What is Collision?)

- Jab do ya do se zyada computers ek hi waqt mein ek hi network cable par data bhejte hain, toh collision (takrao) hota hai.
- Aesa hone se data aapas mein mix ho jata hai, message kharab (corrupted) ho jata hai, aur isay dobara bhejna parta hai.

Bus Topology Mein Collisions Kyun Hote Hain?

- Bus topology mein tamam devices data bhejne ke liye ek hi main cable (jise backbone kehte hain) ka istemal karti hain.
- Kyunke bohat se computers ek hi cable share kar rahe hote hain.
- Is liye, agar do devices ek sath data bhej dein toh unke signals aapas mein takra jate hain.
- **Misal:** Agar Computer A aur Computer B ek hi waqt mein ek hi cable par message bhejain, toh collision ho jayega.

Bus Topology Kya Hai?

- Is topology mein tamam devices ek hi main cable (backbone) ke sath juri hoti hain.
- **Real-life Misal:** Gali mein guzarti hui Cable TV ki taar, jahan har ghar ek hi main taar se connect hota hai. Agar main taar kat jaye, toh sab ka signal chala jata hai.

Faidey (Advantages):

- Yeh sasti (cheap) hoti hai.
- Chote networks ke liye isay lagana bohat asaan (simple) hai.
- Isme cable ka istemal kam hota hai.

Nuqsanat (Disadvantages):

- Agar main cable toot jaye toh poora network fail ho jata hai.
- Isme connect hone wali devices ki tadad mehdood (limited) hoti hai.
- Iski security bohat kam hoti hai.

CSMA/CD Kaise Kaam Karta Hai?

- **CSMA/CD** ka matlab hai Carrier Sense Multiple Access / Collision Detection. Yeh ek tareeqa (protocol) hai jo bus networks mein collisions ko rokne ya handle karne ke liye istemal hota hai.

Kaam Karne Ke Steps:

- **Carrier Sense:** Computer data bhejne se pehle check karta hai ke kya cable par koi aur pehle se data toh nahi bhej raha.
- Agar cable free ho, toh computer apna data bhej deta hai.
- **Collision Detection:** Agar do computers ghalti se ek hi waqt mein data bhej dein, toh system is takrao ko detect kar leta hai.
- **Stop Transmission:** Collision detect hote hi dono computers data bhejna band kar dete hain.
- **Random Wait Time:** Har computer kisi muqarar waqt ke bajaye ek random (be-tarteeb) waqt ke liye intezar karta hai.
- **Resend Data:** Intezar ka waqt khatam hone ke baad, woh apna data dobara bhejte hain.

CSMA/CD Ki Khoobiyan Aur Khamiyan (Features & Limitations)

Khoobiyan (Features):

- **Collision Detection:** Agar do devices ek waqt mein data bhejain toh yeh pata laga leta hai.
- **Carrier Sensing:** Data bhejne se pehle device cable ko check karti hai.
- **Retransmission:** Agar collision ho jaye, toh wait

karne ke baad data ko dobara bheja jata hai.

- **Chote Networks ke liye Behtar:** Jab devices kam hon, toh yeh bohat achay se kaam karta hai.
- Yeh ek asaan tareeqa hai jo bus ya shuruwati Ethernet networks mein asaani se lagaya ja sakta hai.

Khamiyan (Limitations):

- Agar network mein users ziada ho jayen toh performance gir jati hai kyunke takrao (collisions) barh jate hain.
- Baar baar collisions hone se network ki speed slow ho jati hai.
- Yeh sirf shared networks mein kaam karta hai; aaj kal ke modern switched Ethernet mein yeh use nahi hota.
- Isme fasla (distance) mehdood hota hai kyunke cable ki lambai collision detect karne ke amal par asar dalti hai.
- Baday (large) networks ke liye yeh tareeqa theek nahi hai.